

The application of telepresence in medicine

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Telemedicine is a wide ranging subject which encompasses teleconsultation, electronic patient records, continuing medical education and public health information systems. Telemedicine will have an impact upon every aspect of health care delivery from remote clinical analysis and consultation between experts based in specialist hospitals, to teleconsultation as a substitute for physical referral between the GP and a hospital, and ultimately the remote delivery of medical monitoring and care into the home. Telepresence, where remote human expertise is transported by communications networks to a remote location, has its own unique place within the spectrum of telemedicine applications. This paper describes some of those applications of medical telepresence into which BT have been researching with various clinical partners.

1. Introduction

Telemedicine has been defined most clearly as 'the use of electronic information and communications technologies to provide and support health care when distance separates the participants' [1]. Applications of telemedicine in remote consultation fall into two broad categories — store and forward of electronic patient-related information for remote diagnosis, and real-time interactive remote consultation between geographically separate clinicians or patient and clinician. Store-and-forward applications are the fastest growing area of commercial telemedical services, for example, the managed radiological case-reporting service offered by Worldcare Ltd [2]. Patient-related images or case reports are transmitted to a remote expert clinician for report back within 6 to 48 hours. Store-and-forward applications are not explored in this paper since the remote expert is not telepresent during a consultation. This paper details the BT trials and pilots which have helped to establish the credentials for real-time medical telepresence. It highlights technical solutions to clinical concerns which can change real-time teleconsultation from an interesting technology for enthusiasts to a potentially important means of making more effective use of limited human resources, leading to improved patient care at acceptable costs.

2. Remote fetal scanning

The remote fetal ultrasound trial [3] was designed to assess the clinical validity of the transmission of real-time fetal ultrasound images over BT's dial-up ISDN30 service. The system was implemented to provide an expert consultant opinion from a central specialist referral centre to the local clinicians in a hospital. Expectant mothers are routinely scanned at 14–18 weeks of pregnancy at their local ante-natal clinic. Up to 5% of scans in the UK have an ambiguity in the image which would nor-

mally result in the mother being referred to one of the UK specialist fetal care centres for a second opinion. During this trial high-quality ultrasound images generated in the ante-natal clinic at St Mary's Hospital on the Isle of Wight were transmitted over the BT ISDN network to Queen Charlotte's Hospital in London, where for the purposes of referral they were viewed by a consulting expert with virtually no delay or degradation in quality. The technical and geographic configuration is shown in Fig 1.

The equipment in the ante-natal clinic at St Mary's consisted of a stand-alone cabinet containing a BT VC2300 videocodec, an ISDN channel inverse multiplexer, video and audio mixers, plus a small floor mounted mobile control console for use next to the scanner. There was also a wall-mounted low-light camera within the scanning room which captured a general view of the patient and sonographer during the scan. This view allowed the remote consultant to correlate the ultrasound image on view with the position and orientation of the ultrasonic transducer held by the sonographer. The control console allowed the sonographer to determine which images were sent to the consultant or how they should be mixed together. The console had a small LCD monitor which displayed the image being transmitted to the consultant so that the sonographer could be confident the configuration was correct. Figure 2 indicates the view most often seen by the remote consultants at Queen Charlotte's.

A number of human interface issues became apparent after installation. At the scanning end of the link, on the Isle of Wight, the audio system had been configured to allow the remote consultant's voice to be broadcast into the scanning room for all present to hear or for private listening by the sonographer. In practice the local consultant or other

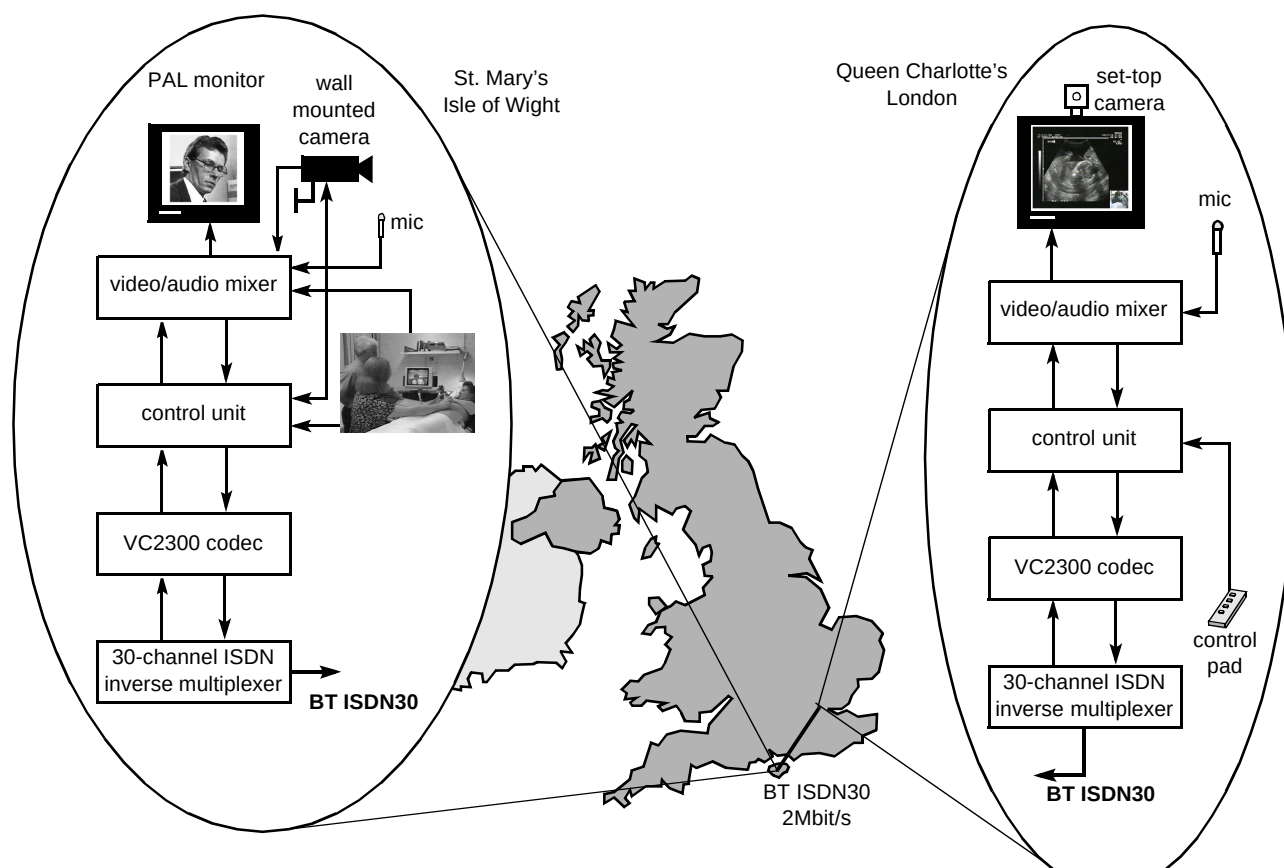


Fig 1 Schematic of remote scanning equipment.

interested clinicians often wished to be present in the room with the sonographer. They required private access to the voice of the remote consultant during their preliminary discussions in order to avoid possible alarm to the expectant mother. This was resolved by the use of cordless infrared audio headsets within the scanning room. The infrared headsets allowed any number of interested parties to hear the remote consultant's comments without introducing variable loading on the electrical audio circuits, and gave the additional benefit of allowing complete freedom of movement for the users.

The remote fetal scanning system was clinically evaluated [4] for a period of 6 months during the pilot and produced the following summary results:

- 39 consultations carried out,
- 86% definitive diagnoses were made over the link,
- 76% abnormalities were detected,
- 4 mothers had to physically attend Queen Charlotte's,
- 80% of mothers felt less anxious using the system.

The four physical referrals were required for more complex tests, not available at St Mary's. Anecdotally the consultants have said that they were as confident in making a primary diagnosis over the system as they were when



Fig 2 Typical ultrasound image with picture in picture.

actually present in the scanning room. The improved quality of life for the mothers should not be overlooked. Mothers, already in a state of anxiety, felt at ease with the system and were happier in the familiar surroundings of their local hospital. At the end of the diagnostic phase of the referral, the sonographer could switch the video input to display the face of the remote consultant, who was able to counsel the mother regarding the outcome of the referral in the same

way as if they were in the same room. Clinician's time was saved at Queen Charlotte's because the evaluation and management advice was provided without having to provide the ancillary support normally required for a referral where the person physically attends.

The remote fetal scanning trial has shown that videoconferencing technology can be packaged to meet the clinical requirements for ante-natal telereferral and to be easy and intuitive to use by the clinicians involved. The use of such a system has enabled referrals to be carried out more quickly by enabling the expert consultant in London to be telepresent in the scanning room of a hospital on the Isle of Wight, which has improved the quality of care provided to expectant mothers during the trial. It has also been suggested that a remote referral system will reduce the number of births of severely handicapped babies and thus reduce the long-term medical and social costs to the state. The higher abnormality detection rate enabled by the remote referral system would reduce the number of medico-legal claims against a hospital Trust. Each successful case against a hospital can cost up to £2M. Nation-wide costs for fetal medicine legal cases are approximately £60M — £100M per annum [5].

Further developments of the system will focus on lower cost terminal equipment and video distribution techniques by which a high-quality videoconferencing gateway could be provided as a shared facility for hospitals, allowing other clinical departments access to their preferred specialist centres for remote consultation. This would increase the system utilisation and therefore make the purchase of the videoconferencing equipment more justifiable for the hospitals.

3. Remote skill acquisition in surgery

The remote fetal scanning trial has shown that videoconferencing systems with standard two-dimensional (2-D) displays can be used to construct telepresence systems suitable for remote medical referrals. However, when telepresence is applied to distance learning where physical manipulative skills are taught, such as in surgery or dentistry, the lack of the third dimension (3-D) or 'depth' in the transmitted images can inhibit the teaching process. To overcome this limitation a 3-D videoconferencing system was developed and tested in surgical teaching. The videoconferencing element of the system was a BT VC2300 and ancillary networking equipment, identical to that used in the remote fetal scanning trial. Two cameras were used instead of one at the image acquisition end of the system and a high-quality 100 Hz video monitor with synchronised LCD shuttered glasses used to display 3-D images at the viewing end of the system. The video signal from each of the cameras was allocated approximately 50% of bandwidth on the 2 Mbit/s ISDN30 circuit linking the two ends of the system.

In the first experiment [6], the system was used to allow a 'trainee' surgeon to be advised by a consultant surgeon who was physically remote from the operation. The trainee surgeon wore a lightweight headset on which two miniature cameras were mounted in order to capture his left and right eye views, as shown in Fig 3.



Fig 3 Twin camera headset.

The cameras were angled relative to each other so that their fields of view converged at the normal point of focus of the surgeon's eyes during the operation. A microphone was integrated within the headset so that the surgeon could speak to the remote consultant. The video signals from the cameras were transmitted, with the audio from the surgeon, through the videoconferencing system, to the 3-D viewing station where the consultant surgeon watched the images on the screen through LCD glasses (see Fig 4).

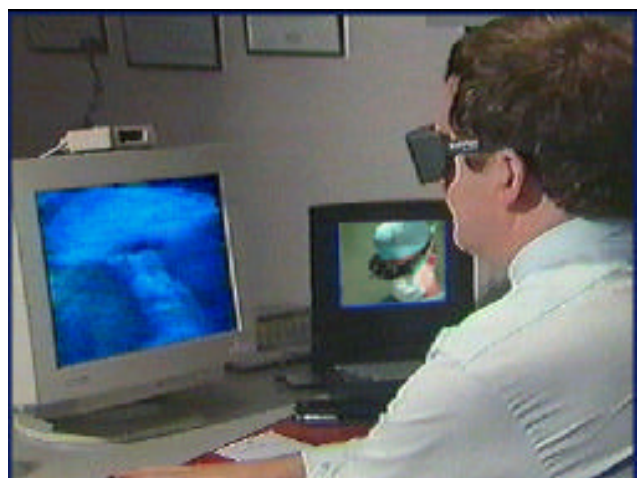


Fig 4 3-D consultant viewing station.

The 3-D system allowed the remote consultant to accurately see, and so comment on, the depth of incisions and the angle at which the surgical tools were being used. The remote consultant surgeon was therefore able to interact with the operating surgeon to give guidance on technique. The 3-D system created a strong sense of shared presence between the surgeons to the extent that the consultant felt he was present at the side of the operating surgeon.

The remote consultant experiment was technically and clinically successful and so a second experiment was designed to test the 3-D system in surgical teaching to larger audiences of medical students. Standard teaching techniques for medical students include watching truncated video sequences of operations recorded by a video camera positioned to one side of the surgeon in the theatre. The usefulness of the 2-D view presented in these videos is limited in the sense that it is not the view seen by the surgeon. To quantify the teaching benefits of real-time 3-D images to large audiences, medical students were invited to view a total hip replacement operation in both 2-D and 3-D video.

The video from the surgeon's headset cameras was transmitted to two distant lecture theatres and presented to a group of 20 students — 2-D video displayed in one theatre, and 3-D video displayed in the other. In both theatres the video was displayed by projection on to a large screen approximately 3 m × 3 m in size. The 3-D video was displayed using a polarising projector, which circularly polarised the left and right eye views, by splitting the views between adjacent horizontal video scan lines. This 3-D video projection technique produced a 3-D image with 50% of the horizontal resolution of the 2-D video image. The students observed the video through polarising glasses. The students were allowed to view the 2-D and 3-D transmissions at will, and were asked to rate their opinions of the quality of each in a structured questionnaire. The operation took two hours during which time an experienced surgeon joined the students to moderate questions to the operating surgeon. The two surgeons discussed working methods and specific details about the operative process over the video-conferencing system.

The results of the questionnaires indicated that the compromise in resolution required for practical large screen 3-D displays impaired the fidelity of 3-D video to the extent that there was no perceptible benefit over standard-resolution 2-D video. However, developments in large screen immersive 3-D displays such as the VisionDome [7], in which large 3-D video images can be viewed without compromising the video resolution, are likely to make 3-D remote surgical training effective in the future. It will then be possible for large groups of medical students to be simultaneously telepresent with the operating surgeon.

4. Telepresence in accident and emergency

Accident and emergency (A&E) medical care is usually delivered by paramedics at the scene of an incident, e.g. a road traffic accident. A telemedicine system has been investigated which enabled other expert clinicians to be telepresent at the accident to support the work of the paramedic, or to plan subsequent medical care. A primary requirement for a telemedical solution for paramedics is that the system be transportable and not impede the free movement of the paramedic around the accident site. CamNet [8], a system previously designed for engineering applications, was tested as a mobile telemedical solution. CamNet is an audiovisual headset system which had been developed to allow the user to interact with a distant expert using both audio, video and graphical information, in order to transport the expert's skills to the user in the field. In addition to carrying a conventional microphone and earphone, the CamNet headset has a small CCD camera and a miniature view screen. The screen is viewed with the right eye at a distance of a few centimetres. The optics inside the view screen creates a display which appears equivalent to that of a standard PC monitor viewed at a distance of 1m from the eye. The image produced in the display is an intense monochrome red on black high resolution dot matrix with a contrast ratio of 70:1, making it ideal for use in both bright and dark ambient light conditions.

The CamNet headset has been used on trials within the medical facility on the Fortes Bravo North Sea oil rig to provide expert medical diagnosis from the mainland to paramedics on the rig. The previous level of support offered to the rig paramedic was via a structured fax questionnaire from the Aberdeen Royal Infirmary. The questionnaire was completed by the paramedic and faxed back to the mainland where a consultant evaluated the information and called the rig to give his opinion. The patient was then either treated on board the oil rig or air lifted as a medivac (medical evacuation). The deployment of CamNet allowed clinicians on the mainland to see, for the first time, the physical condition of the remote patient which enabled the clinicians to make a more informed decision about the management of the patient. In one example of its use, CamNet enabled clinicians on the mainland to give instruction to a paramedic on how to stabilise a patient who had suffered a head injury.

The successful trials of the CamNet system on the oil rig were followed by a one-day event in which the crew of an East Anglian ambulance were equipped with CamNet to evaluate its use in a simulated road traffic accident [9]. The CamNet system was deployed with the ambulance using a portable Inmarsat 64 kbit/s dial-up satellite communications unit to provide mobile access to the BT ISDN2 network. The connection was made to a BT VC7000 desktop videoconferencing system in the Accident and Emergency

Department of Ipswich Hospital. The paramedic wore the CamNet headset and belt pack, which communicated via a commercial digital radio local area network to the Inmarsat terminal in the ambulance vehicle (Fig 5).

The information transmitted to the consultant was found to be useful in establishing a fast and accurate diagnosis of the casualty's injuries. By viewing the received video from the CamNet headset on the VC7000 videoconferencing monitor, and by transmitting graphics of the structure of the human chest to the paramedic, the A&E consultant at Ipswich hospital was able to guide the paramedic through the simulated procedure of inserting a lung drain into the casualty. The trial of CamNet showed that telepresence technology can be successfully used to speed the delivery of A&E care by linking the roadside paramedic and the hospital-based consultant, but the present cost of CamNet makes it appropriate for use only in major incidents.



Fig 5 Paramedic wearing CamNet.

Capitalising on the experience gained from the CamNet trial, BT Laboratories, in collaboration with Apple Computer, developed and tested TraumaLink [10]. TraumaLink is a low-cost system that combines digital still imaging

and GSM cellular network technologies to transfer still images from paramedics at the scene of an incident to an A&E department (see Fig 6). The images enable the A&E staff to be forewarned of the nature of the accident and of the likely treatments the casualty will require on arrival at the hospital.

TraumaLink consists of a small fixed focus digital still image camera with integrated flash unit, a dash mounted cradle for the camera, a notebook PC and a GSM cellular phone. The camera is used by the ambulance technician whilst the paramedic carries out normal patient care procedures. After capturing pictures of the incident and the position of the casualty, the technician places the camera into the cradle. The technician has no other operations to perform to operate the system. The images are automatically compressed and downloaded into the on-board PC. The PC initiates a GSM data call at 9.6 kbit/s from the ambulance to the receiving PC in the hospital A&E department. Each full-colour image requires approximately 30 seconds transmission time. Clinical A&E staff in the hospital view the received images on a PC workstation and make colour prints for reference in the treatment bay. The TraumaLink system, which has been designed to be as simple and intuitive to use as possible, is being evaluated by the Norfolk and Norwich hospital.

5. Conclusions

Telepresence has proved to be clinically effective in specific aspects of medical care, particularly in the provision of remote specialist consultation where real-time interaction is required between the consultant and the clinician attending a patient. The acceptability of telepresence in the delivery of health care is highly dependent on the simplicity of the user interface and on ensuring that the technology is properly integrated with other clinical systems. It should also be recognised that clinicians will not change working practices unless clear benefits can be demonstrated in terms of improved patient care. In addition, health care managers expect demonstrable

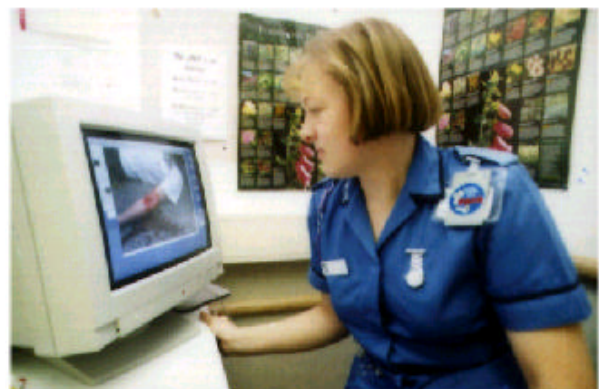
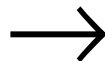


Fig 6 Traumalink image capture and display.

efficiency gains from new systems, particularly where new technology requires widespread change in working practices and hence investment in change-management training and support.

Many successful pilots of the use of telepresence in medicine are ongoing in the UK in both urban [11] and rural environments. One approach to implementing telemedicine is multilayered applications [12], using store-and-forward technology initially with real-time telepresence options being reserved for difficult cases. In truly remote areas telepresence is growing in acceptance [13] as a means of facilitating the delivery of health care. As hardware and communications technologies continue to reduce delivery costs, health services will be revolutionised by telemedicine applications that use telepresence. This leads to the prospect of world-class health care that is available to all, irrespective of their location or ability to travel.

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In 1993 he moved to the telemedicine research team at BT Laboratories, where he was specifically responsible for managing the BT telemedicine trials. Since 1996 he has led the Telematics group which is concerned with research into both Telemedicine applications and the related emerging area of telecare.

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Melanie Collins joined BT Laboratories in 1987. She initially worked in the team producing submarine receivers. She then moved on to configurational research into a new high-power fibre amplifier.

In 1994 she began working in the field of telemedicine. She now heads the healthcare research aspect of the telecare project and also has the responsibility of designing, presenting and disseminating associated project information.



Steve Webster has a degree in Communications Engineering from the Open University. Since joining BT in 1981 he has worked in many different research areas covering submarine cable systems, advanced optoelectronic devices and advanced telepresence systems. While working on advanced optoelectronic devices he spent a period of secondment at BT&D. More recently he has been involved in research activities looking at network facilitated health care delivery and co-ordination. Currently these telemedicine activities involve the delivery of remote consultations and

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Duncan Rose joined BT Laboratories in 1988 with a degree in Electronic Engineering. He worked initially on data encryption systems, high efficiency modulator and demodulator DSP coding techniques and developed ultra-high-speed circuitry for satellite systems.

Since 1993 he has investigated the application of signal processing techniques for the benefit of hearing-impaired telephony users and developed telepresence systems to increase the realism of videoconferencing systems. During the past 18 months he has developed and managed various telemedicine

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